

Title: Racial and Gender Differences in Employment and Earnings in the United States (2015–2020)

Research Question: Do median weekly earnings differ by race and gender in the United States from 2015 to 2020?

- The response variable is median weekly earnings, which measures the typical weekly income earned by individuals.
- Important covariates include race, sex, age group, and year.
- By analyzing these variables, we hope to understand how earnings vary across demographic groups and whether these differences change over time.

Description: The data comes from the U.S. Bureau of Labor Statistics (BLS), compiled for the TidyTuesday project on February 23, 2021. The earnings data were sourced from the BLS Current Population Survey. The `earn.csv` file contains 20,832 observations of quarterly earnings data from 2010–2020. Key variables include `median_weekly_earn`, `sex`, `race`, `age`, `year`, `quarter`, and `n_persons`.

Our proposed response variable will be a binary variable indicating a high earning status. The main predictors in our model will be sex, race, and age group, which are all categorical variables. The year variable will also be grouped making it categorical, helping us understand how trends change over time.

These predictors were chosen because salary and earnings often vary between demographics. Our model will help us identify and understand these trends over time.

Citation: TidyTuesday. (2021, February 23). *Employed and Earn Status*. Retrieved from <https://github.com/rfordatascience/tidytuesday/blob/main/data/2021/2021-02-23/readme.md>. Original data source: U.S. Bureau of Labor Statistics.

Proposed methods: To address our research question regarding racial and gender differences in earnings, we will transition our response variable from a continuous dollar amount to a binary categorical variable. We will define our response, “High-Earner Status,” by setting a threshold at the overall median weekly wage across all groups (approximately \$900 - \$1000). Observations at or above this threshold will be coded as 1, and those below as 0.

The primary model for this analysis will be a logistic regression model. This is the appropriate method for this course because it allows us to model the log-odds of being a “high earner” as a function of our key covariates: race, sex, age group, and year. By using a logistic model, we can calculate the odds ratios to quantify, for example, how

much more likely one demographic group is to exceed the median income threshold compared to another. We may briefly contrast these findings with a standard linear regression (predicting raw earnings) to show the difference in interpretation, but the logistic model will serve as the central framework for our project.